

UNIVERSITY OF SCIENCE

FACULTY OF INFORMATION TECHNOLOGY | SOFTWARE ENGINEERING DEPARTMENT

INTRODUCTION TO SOFTWARE ENGINEERING

PROJECT ASSIGNMENT 0

Project Idea & Team Contract

HomeMate AI

AI-assisted coordination for self-managed residential construction

DOCUMENT INFORMATION	DETAILS
Team ID	T08
Team name	Không mười điểm thì bị cười
Team lead	Huỳnh Minh Hùng (24127385)
Document status	Draft for team confirmation and TA approval
Submission date	13/10/2026

Teaching Assistant

Trần Duy Hoàng

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Objectives

This Project Assignment 0 (PA0) document records the proposed project idea and the working agreements of Team T08. It is intended to support the teaching assistant's review before the team proceeds to PA1.

- Present the project concept, intended users, initial feature priorities, feasibility considerations, and questions for teaching-assistant feedback.
- Define team responsibilities, communication arrangements, quality controls, decision procedures, and methods for resolving issues.
- Document the team's agreed approach to generative AI and maintain an AI usage log in accordance with course policy.

1 Team Information

ITEM	VALUE
Team ID	T08
Team name	Không mười điểm thì bị cười
Team lead	Huỳnh Minh Hùng — 24127385
Primary communication channel	Microsoft Teams — NMCNPM / Nhóm 8 / General

STUDENT ID	FULL NAME	INSTITUTIONAL EMAIL	MAIN ROLE	BACKUP ROLE
24127385	Huỳnh Minh Hùng	hmhung2427@clc.fitus.edu.vn	Team Lead / PM	Designer
24127464	Nguyễn Phạm Minh Ngọc	npmngoc2413@clc.fitus.edu.vn	Business Analyst	Tester
24127515	Phạm Minh Quân	pmquan2435@clc.fitus.edu.vn	Designer	Developer
21127061	Trần Gia Huy	tghuy211@clc.fitus.edu.vn	Developer	Business Analyst
24127570	Võ Thúc Trí	vttri2418@clc.fitus.edu.vn	Tester	Team Lead / PM

If the above Microsoft Team link is inaccessible:

https://teams.microsoft.com/l/team/19%3AIKCEIHH2K1p3QK1_suVJdIw86xEpTIyvjXCj63jo_2k1%40thread.tacv2/conversations?groupId=bdfca8d4-3adf-489f-bf18-7db474df6c85&tenantId=40127cd4-45f3-49a3-b05d-315a43a9f033

2 Project Idea Overview

2.1 Project Name and One-Sentence Pitch

HomeMate AI is a proposed web application that helps Vietnamese households coordinate self-managed home construction or renovation by tracking expenses and workers, maintaining an editable construction-stage schedule, and obtaining context-aware AI assistance.

2.2 Problem and Motivation

Some households manage residential construction directly rather than purchasing a fully managed construction service. They purchase materials, engage workers, and monitor work themselves, often using informal records and conversations. This arrangement requires coordination across budgets, workforce attendance, work sequencing, and on-site evidence.

For owners without specialist construction experience, fragmented records can make cost overruns, delayed activities, and gaps in progress documentation difficult to identify. General-purpose expense trackers and chat assistants do not necessarily organize these records around the phases of a household construction project. HomeMate AI proposes to integrate them into a single project-oriented workflow.

The application would support informed monitoring and planning, not replace licensed architects, structural engineers, site supervisors, or professional inspections. AI-generated technical content must be treated as preliminary guidance and checked against applicable standards and qualified advice.

2.3 Target Users

USER GROUP	PRIMARY NEEDS
Household owners / families	Create or adjust a stage-based work plan; record material and labour costs; review workforce totals and progress; ask contextual questions; optionally retain receipts and site photos.
Lead workers / crew coordinators	Review assigned work or expected daily tasks; communicate progress; report worker presence and quantities, subject to permissions granted by the owner.

2.4 Main Features

The following seven features are initial priorities rather than a finalized requirements baseline. “Must-have” features define the implementation target for PA5; optional features may be deferred.

#	FEATURE	INITIAL DESCRIPTION	MUST-HAVE
1	Cost and material records	Log budget categories, purchases, quantities, labour expenditure, and basic budget summaries.	Yes
2	AI-assisted construction-stage planning	Generate an editable high-level sequence or schedule from owner-entered project details; record basic stage progress.	Yes
3	Stage checklists and technical reminders	Present phase-relevant checklists using reviewed sources and clearly identified limitations.	No
4	Context-aware AI question answering	Answer general planning and record-related questions using the project’s provided context, with safety disclaimers.	Yes
5	Worker attendance and wage estimates	Record daily attendance or working days, agreed rates, and estimated amounts owed.	Yes
6	Construction evidence archive	Attach photos, receipts, and notes to dates or construction stages.	No
7	Exception notifications	Highlight user-defined budget thresholds or delayed activities.	No

2.5 Similar Products

Illustrative points of comparison include Money Lover (general personal expense tracking), MISA software (accounting and financial management), and ChatGPT (general-purpose conversational assistance). Unlike any one of these categories in isolation, HomeMate AI would connect residential construction records, work-stage tracking, and project-context questions in a homeowner-oriented interface.[1]

[1] This is an initial positioning hypothesis; a documented competitor review is planned for PA1.

2.6 Platform and Technology Ideas

Initial platform: Responsive web application; native mobile support may be considered later, but is not included in the initial delivery commitment.

Potential stack: React and Tailwind CSS for the interface; Supabase with PostgreSQL for authentication, data, and managed back-end services; an external language-model API (e.g., OpenAI or Google Gemini) for AI-assisted features. Provider selection remains open.

Information handling: Use structured project records and bounded conversation context. The feasibility of retrieval, summarisation, image inputs, API costs, and privacy controls requires further evaluation.

2.7 Feasibility, Scope and Open Questions

Feasibility assessment. A five-person team may be able to implement a constrained prototype during the approximately three-week implementation window (weeks 8–10), provided that requirements and design are agreed in PA1–PA4, managed services reduce infrastructure work, and the AI features use a narrow, explicitly defined scope. Full building-plan interpretation or dependable engineering advice is not a credible commitment for this schedule.

Proposed MVP boundary: Manual entry of construction details; itemised costs; a simple user-editable stage schedule; worker attendance and wage calculations; and basic context-aware AI chat. The AI may propose a plan but cannot independently validate structural safety or legal compliance.

Out of scope: Professional structural calculations, certified technical recommendations, detailed architectural drafting or 2D/3D modelling, interior design, feng shui optimisation, supplier marketplaces, and worker recruitment or payments.

KEY RISK	INITIAL RESPONSE
Unreliable technical AI responses	Restrict answers to general information; communicate uncertainty; require qualified professional verification for safety-critical choices.
Excessive feature scope and short coding period	Implement must-have workflows first, use explicit acceptance tests, and defer photo storage, notifications, and advanced analysis.
Long or sensitive project histories; API cost	Keep structured records in the database, send only needed context to the model, and evaluate usage limits and data minimisation.
Incomplete or inconsistent user records	Provide guided inputs, validation, and editable schedules rather than implying automated precision.

Questions for the TA: (1) Is the proposed Must-have set sufficiently narrow for PA5? (2) What evidence or source restrictions should be required for AI-generated construction guidance? (3) Would lightweight project-record retrieval or summarisation be an acceptable strategy for maintaining AI context within the course schedule?

3 Team Contract

This agreement defines how members will collaborate and document contributions during the course. Any changes to the commitments below should be discussed, recorded, and acknowledged by the team.

3.1 Goals and Expectations

The team aims to achieve the highest attainable assessment level while meeting the course's documentation, implementation, testing, and teamwork criteria. Each member plans to contribute at least six hours per week, typically approximately one to two hours on active working days. Time commitments may vary with other academic responsibilities, but members must communicate foreseeable capacity constraints early.

3.2 Communication

Primary channel: Microsoft Teams, General channel[2]; direct messages are available for urgent coordination.

Response standard: Members aim to acknowledge messages within 24 hours, permits up to 72 hours in non-urgent circumstances. Work-blocking or near-deadline matters should be flagged directly and escalated without waiting for the maximum response period.

Decision records: Relevant agreements, action items, and unresolved issues are recorded on the task board or in meeting notes; source code and deliverables are retained in the Git repository.

SHARED RESOURCE	EDITABLE REFERENCE
Jira task board	Link https://clc-team-dj4m9b5b.atlassian.net/jira/software/projects/SCRUM/summary?atlOrigin=eyJpIjoiMzBlYmI2OWE5M2Y0NDc4MTgxZDA5M2EzNTE5Y2I0ZjMiLCJwIjoiaiJ9
GitHub repository	Link https://github.com/AdrianParry-17/HomeMate-AI
Google Drive / shared documents	Link https://drive.google.com/drive/folders/1LJxy-mKmLc8_x1GGMaSoO3hkU3QZ41d-?usp=sharing

3.3 Meetings

TA status meeting: Sunday, 19:30, online

Internal team meeting: Sunday, 09:30; rescheduling is permitted when agreed in advance.

Attendance: Members inform the team before an absence where possible; absences and follow-up actions are recorded in the weekly status report.

Weekly status report: Sunday at 00:00

[2] Link to the Team provided in Section 1, look for the General channel

3.4 Roles and Responsibilities

Primary and backup roles are identified in Section 1. The following task ownerships reflect the team's initial allocation; additional owners should be assigned during PA1 as the work breakdown and technical architecture are established.

RESPONSIBILITY	INITIAL OWNER
Project reporting and coordination	Huỳnh Minh Hùng
Business analysis and requirements writing	Nguyễn Phạm Minh Ngọc
Presentation slides	Võ Thúc Trí
UI/UX design	Phạm Minh Quân
Software structure and architecture design	Huỳnh Minh Hùng
Frontend implementation	Huỳnh Minh Hùng
Backend implementation	Trần Gia Huy
AI integration	Trần Gia Huy
Testing coordination	Võ Thúc Trí
Repository protection, CI and deployment ownership	Huỳnh Minh Hùng[3]

3.5 Work Standards

Deadlines: Each task has an intended completion date and a final internal deadline set early enough to protect the related PA submission. The task owner informs the team immediately when either is at risk.

Quality assurance: Each substantive document section and pull request must be reviewed by at least one other member. Review comments and corrections should be traceable to the revised work.

Task management: Project work is recorded on the team board with an owner, due date, and status. Small or urgent tasks may be raised by direct message, but should subsequently be reflected in the board where material.

Version control: Development takes place on topic branches, and changes to the protected main branch are merged through reviewed pull requests. Direct pushes to main are not permitted, consistent with the course rubric.

Weekly reports: The team lead or backup submits the agreed report each week, commits it to /docs/management/weekly/W<week>-status.docx (or .md), and does not rewrite prior weeks' reports.

3.6 Use of AI Tools

AI tools may support research, drafting, coding, and review, but use must be disclosed, verified by a responsible team member, and compliant with the course academic integrity and confidentiality rules.

TOOL CATEGORY	PERMITTED USE	HUMAN REVIEW REQUIREMENT
Conversational AI (e.g., ChatGPT, Gemini, Claude)	Brainstorming, explanations, outline planning and language editing.	Check factual and technical claims against course material and reliable sources; edit and approve before submission.
Coding agents (e.g., Copilot, Codex, Claude Code)	Implementation suggestions, refactoring, test scaffolds and troubleshooting.	Understand all submitted code, inspect changes, run appropriate tests, and use the standard review process.
Other tools if adopted	Use only after the team records their role and intended outputs.	Apply equivalent disclosure, human verification and confidentiality controls.

AI usage log custodian: Huỳnh Minh Hùng (team lead). Each member is responsible for reporting significant AI-assisted work and explaining the resulting deliverables.

Confidentiality: Do not enter personal or confidential project information into external AI tools. AI-generated construction guidance must not be presented as certified engineering advice.

3.7 Decision Making

The source team agreement calls for more than three affirmative votes among five members (that is, at least four) for a substantive team decision. Members should first discuss alternatives and document opposing views. If the required agreement is not reached, the decision may be deferred when feasible; for time-sensitive operational matters, the team lead

[3] Could be changed in PA1.

may decide after considering the members' positions and documenting the rationale. The team should reconfirm this four-of-five threshold before signing, because it is stricter than a simple majority.

3.8 Handling Conflicts and Missed Work

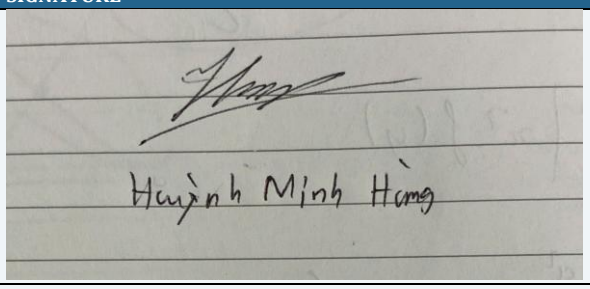
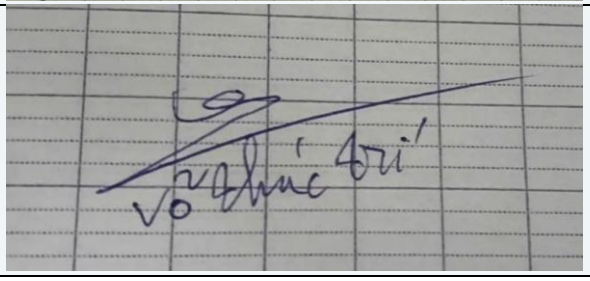
Step 1 — Direct resolution: Discuss the concern privately with the member involved, agree on a corrective action and revised date, and identify any impact on dependent tasks.

Step 2 — Team review: If unresolved or recurrent, raise the issue at a team meeting, distribute work as necessary, and record decisions and follow-up in the weekly status report.

Step 3 — TA escalation: If the issue materially affects delivery or individual contributions, notify the TA with factual evidence from task records, review history, and status reports.

3.9 Signatures

By signing below, each member acknowledges the responsibilities, procedures, and expectations in this contract. To insert a scanned or electronic signature in Word, click the signature cell, remove its placeholder text, and insert the image inline; then enter the signature date.

STUDENT ID	FULL NAME	SIGNATURE	DATE
24127385	Huỳnh Minh Hùng		*13/10/2026*
24127464	Nguyễn Phạm Minh Ngọc	*Ngọc*	*13/10/2026*
24127515	Phạm Minh Quân		*13/10/2026*
21127061	Trần Gia Huy		*13/10/2026*
24127570	Võ Thúc Trí		*13/10/2026*

4 AI Usage Log

The following entry records the AI-supported of this PA0 document.

#	MEMBER	TOOL	TASK / PURPOSE	HOW OUTPUT WAS VERIFIED AND CHANGED
1	Nguyễn Phạm Minh Ngọc	ChatGPT	Reorganized, refinement and extension of the original project ideas that used as references for this document.	Verified, fact check, by both Nguyễn Phạm Minh Ngọc, Huỳnh Minh Hùng, and other team members at the initial meeting.
2	Huỳnh Minh Hùng	ChatGPT	Reorganised and edited the PA0 draft for academic clarity, consistency, template alignment, and Word/PDF formatting from the original Markdown.	Verified, fact check, changed and re-refinement by Huỳnh Minh Hùng (Team Lead)